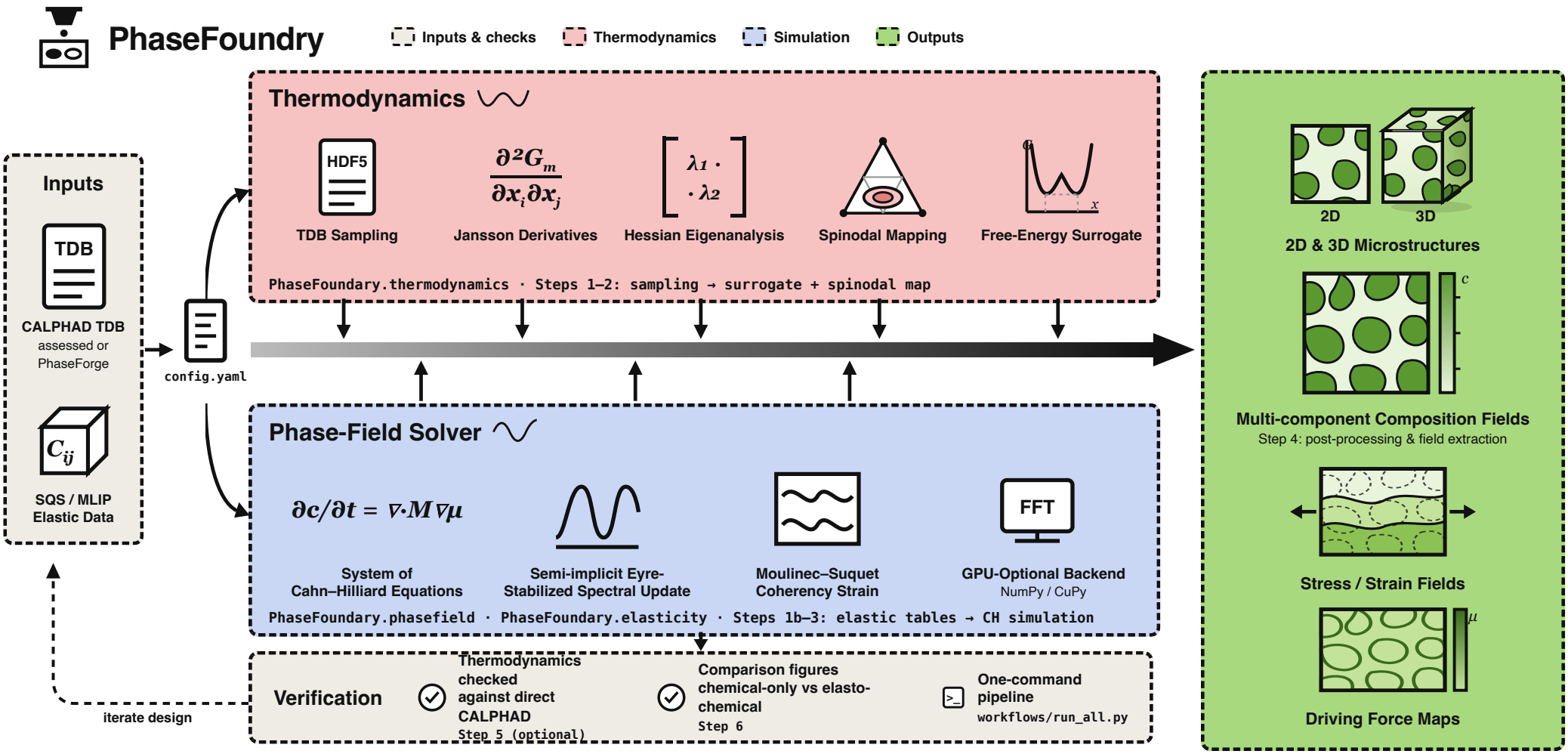


Fig. 1 | Workflow of *PhaseFoundry*. From a CALPHAD TDB (assessed or MLIP-generated via PhaseForge), SQS/MLIP elastic data, and a single YAML configuration, PhaseFoundry samples the database with analytic Jansson derivatives, maps spinodal regions through gauge-invariant Hessian eigenanalysis, fits a free-energy surrogate, and evolves the instability with an Eyre-stabilized pseudo-spectral Cahn–Hilliard solver with optional Moulinec–Suquet coherency strain, producing spinodal microstructures, composition fields, and stress/strain fields. Verification stages (Steps 5–6) validate tables and derivatives against direct pycalphad calculations and compare chemical against elasto-chemical simulations.